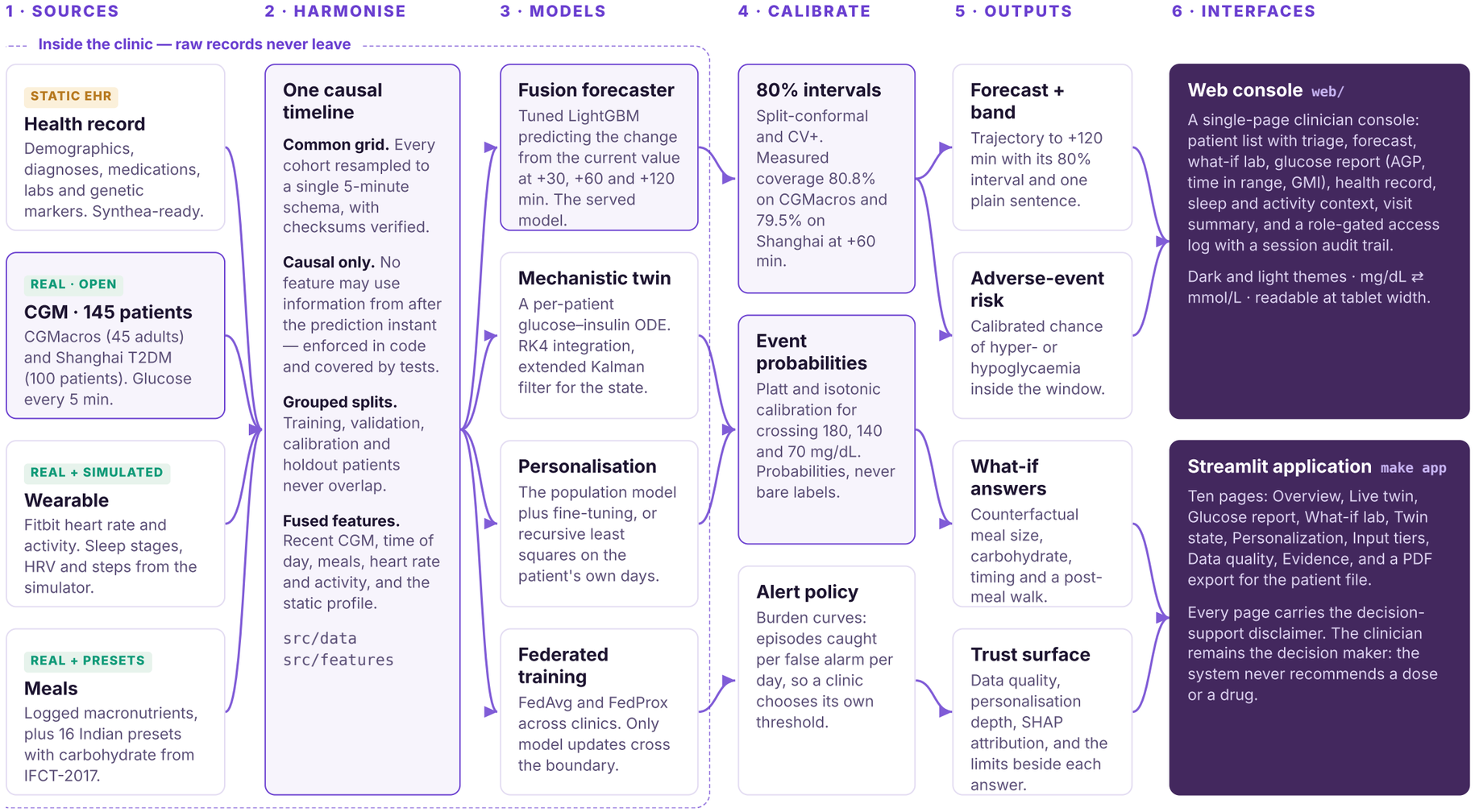
GlucoTwin — system architecture

A patient-specific type 2 diabetes digital twin. Four input streams are harmonised onto one causal timeline, fused by the model layer, calibrated into intervals and event probabilities, and served to two clinician interfaces. Raw records never leave the clinic.

Happiest Health Digital Twin Challenge — Build. Predict. Reimagine Healthcare.



METHOD — HOW EVERY NUMBER WAS OBTAINED

- **Pre-registered.** Each protocol and decision rule written to `docs/DECISIONS.md` before the run that tested it. Over 70 entries.
- **Paired patient bootstrap.** Confidence intervals computed over patients, never over time windows — readings from one patient are not independent.
- **Untouched holdout.** Final claims confirmed on patients never used for any choice anywhere in the project.
- **Null results published.** Where a signal does not help, the number is reported at full size rather than dropped.

MEASURED RESULTS

METRIC	VALUE
MAE at +30 min	11.6
MAE at +60 min	16.8
MAE at +120 min	21.2
AUROC, above 180	0.82
AUROC, below 70	0.81
Interval coverage	80.8%

PRIVACY BY DESIGN

- Anonymised open or synthetic data only. No real patient record enters the repository.
- **Federated:** FedAvg matches pooled central training (difference 0.09 to 0.12 mg/dL, intervals include zero) and beats a single small clinic alone by about 2 mg/dL.
- Role-gated clinician access with a session access log.
- Data handling mapped to India's **DPDP Act** in `docs/limitations.md`.

SCOPE AND LIMITS

- One condition: **type 2 diabetes**.
- Decision support only — never a dose or a drug. Not a medical device.
- The India, sleep and federated results are **simulations**: they test design, not clinical performance.
- Not prospectively validated.

Code MIT. Datasets keep their own licences. Simulated streams are labelled SIMULATED throughout.